

AIRA JOSH C. YNTE

CURRICULUM VITAE

Computer Engineer

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PROFESSIONAL SUMMARY

Detail-oriented Computer Engineering graduate with hands-on experience in software development, embedded systems, and full-stack web applications. Passionate about designing robust systems, bridging the gap between hardware and software, and building performant user-facing applications. Skilled in React, TypeScript, C/C++, Flutter, and system architecture.

EDUCATION

Bachelor of Science in Computer Engineering (Magna Cum Laude)

2022 – July 2026

Colegio de Muntinlupa — Muntinlupa City, Metro Manila, Philippines

Specialization: Software Architecture & System Design

- Core coursework: Software engineering, user interface design, database systems, computer network structures, system simulation.
- Focused projects on real-time data visualization, client-side state machine integration, and responsive layout designs.
- Designed a muscle bio-signal telemetry web controller on the ESP32, translating hardware signals into fluid canvas charts.

PROFESSIONAL EXPERIENCE

Freelance Software Engineer

2023 – Present

Various Clients

- Designed and delivered 5+ high-performance web applications using React/Vite and TypeScript, reducing client-side initial load times by 35%.
- Integrated interactive Recharts telemetry data dashboards, plotting 10,000+ data points with interactive query latency under 100ms.
- Built secure client-side Web3 transaction signing modules using EIP-1193 MetaMask standards, protecting user asset security.
- Collaborated with business stakeholders to translate 30+ Figma wireframes into responsive, accessible interfaces (WCAG compliant).
- Optimized rendering cycles, state management, and asset sizes to reduce average initial page load times by 40%.
- Managed 10+ client review cycles, implementing quick frontend enhancements and carrying out successful production handovers.

IC Design Intern (Design & Verification)

2025

Xinyx Design Consultancy and Services Inc.

- Designed and verified AMBA APB3 peripheral interfaces using SystemVerilog, achieving 100% code coverage across 20+ simulation test cases.

- Gained hands-on experience mapping logic layouts and verifying timing constraints in behavior simulations, analyzing 50+ waveforms and log profiles.

PROJECTS

AeroVit — Gamified AI Fitness Platform (Thesis Project)

Awarded Best Thesis

2024 – 2026

Role: Lead Developer & Embedded Systems Architect

- Designed and built a custom ESP32-S3 smartwatch running C++/FreeRTOS firmware to stream HR and IMU metrics over BLE at 50Hz with 98% packet reliability.
- Developed a Flutter mobile application integrated with MediaPipe BlazePose (33 landmarks) for real-time pose estimation at 30+ FPS.
- Implemented turn-based dungeon gamification (Flame 2D) and on-chain ERC-20 reward withdrawals (Solidity) with a 5% on-chain burn on Sepolia.

Fehuvia — Open Finance B2B SME Treasury Workstation

2026

Role: Lead Web3 & Full-Stack Developer

B2B treasury workstation featuring a custom glassmorphic interface, interactive cashflow metrics, AI invoice digitization drawer, and a browser-side cryptocurrency transaction panel.

- Designed a responsive glassmorphic dashboard in React 19 and Tailwind CSS, reducing average obligation load times by 40%.
- Programmed client-side Web3 transaction signing via EIP-1193 MetaMask standards to support secure settlements in under 2 seconds.
- Built an Express.js/PostgreSQL backend with an event listener daemon to synchronize L2 event logs asynchronously with 99.9% database integrity.

StickOut — Interactive VLSI EDA Editor

2026

Role: Creator & Lead Frontend Engineer

- Developed an interactive EDA tool with an infinite 2D HTML5 canvas viewport, rendering and redrawing layout graphics smoothly at 60 FPS.
- Formulated an auto-bridging intersection algorithm to automatically resolve short-circuits on same-layer wire crossovers in under 5ms.
- Programmed a custom double-pass regex parser and canvas rendering engine to dynamically parse and draw LaTeX subscripts (e.g. V_{DD}) in real-time.

Plant.io — 8-Bit IoT Nursery Dashboard

2025

Role: Full-Stack Mobile & IoT Developer · Freelance

Cross-platform Flutter dashboard featuring a Y2K/8-bit retro dark terminal aesthetic, real-time sensor plots, and plant-health scoring.

- Created a Y2K/8-bit retro dark-mode Flutter interface pairing with ESP32 microcontrollers for real-time sensor telemetry.
- Engineered remote actuator control channels (pumps, fans, fertilizer dispensing) via Firebase Realtime Database.
- Implemented offline data caching, connectivity monitoring, local push alerts, and interactive `fl_chart` sensor trends.

The Safehouse — Community Showcase & Pitch Site

2025 – Present

Role: Frontend & 3D Web Developer

- Built an interactive 3D valley journey scene in React Three Fiber with procedural terrain generation and camera path tracking.
- Wired GSAP timelines and ScrollTrigger to Lenis smooth scroll, choreographing 13 narrative scroll sections.
- Implemented accessibility-first MotionModeContext to respect user motion preferences, providing static CSS fallbacks.
- Integrated interactive Rive vector animations, split-heading reveals, and parallax components for narrative storytelling.

Manhwa Reader — Web Application

2026

Role: Frontend Developer · Personal Project

- Created a single-page web application in React 19, TypeScript, and Tailwind CSS to browse and read manga via MangaDex API.
- Designed custom dark/light color themes with local storage persistence and fluid transitions.
- Formulated defensive UI states (loading spinners, empty slots, error notifications) to handle inconsistent API payloads.
- Deduplicated chapter lists and optimized image loading times, selecting chapters with rich page coverage dynamically.

Student Consultation System

2025

Role: Full-Stack Developer · Freelance

- Designed role-based dashboard layouts in React/Vite supporting distinct interactive views for students, professors, and admins.
- Formulated a highly interactive consultation scheduling calendar, featuring custom grid structures and drag-and-drop actions.
- Built the user interface for real-time messaging, including attachments drawers, image crop-and-previews, and status indicators.
- Implemented client-side routing, protected navigation guards, and global user-profile state using React context.

Walang Basagan ng Thrift — E-Commerce Platform

2025

Role: Full-Stack Web Developer · Freelance

- Created Y2K storefront design in React 19/Tailwind, incorporating rounded container grids and a custom pink-white palette.
- Built interactive shopping cart drawers, catalog filter bars, and modal-based order history reports.
- Coded admin dashboard for product CRUD, user list moderation, order tracking, and customer support threads.

SKILLS & COMPETENCIES

Software Development: Clean architecture, API engineering, full-stack debugging, performance optimization, modular system design

System Design & Architecture: Modular architecture planning, database schema design & planning, client-server models, decentralized system protocols

Embedded & Hardware Engineering: Firmware development, real-time operating systems (RTOS), digital logic verification (AMBA), PCB layout & prototyping

Design & Interactivity: UI/UX wireframing, high-fidelity canvas viewport rendering, micro-interactions, motion choreography & narrative layout

Professional Capabilities: Technical documentation & spec writing, cross-functional collaboration, client communication & iterative feedback, rapid upskilling

Tech Stack Summary: TypeScript, JavaScript, C/C++, Python, SQL (PostgreSQL, MySQL, SQLite), React, Next.js, Express, Node.js, Flutter, Tailwind CSS, Solidity, SystemVerilog, Git, Docker, KiCad, Figma, Firebase, MongoDB

CERTIFICATIONS & TRAINING

- **AI Fluency: Framework & Foundations** — Anthropic
- **Introduction to Model Context Protocol** — Anthropic
- **Model Context Protocol: Advanced Topics** — Anthropic
- **The AI Engineer Path Certificate** — Scrimba
- **On-the-Job Training Certificate** — Xinyx Design Engineering, Inc.
- **Blockchain4Youth Certificate** — Bitget Blockchain4Youth
- **Learn React Certificate** — Scrimba
- **Learn TailwindCSS Certificate** — Scrimba
- **Zuitt Game Dev Certificate** — Zuitt
- **TINA Design Suite Workshop** — Hytec Power & Inc.
- **Understanding EDR: Protecting Endpoints Against Modern Threats** — Cybersecurity
- **Introduction to Git** — DataCamp
- **Intermediate Git** — DataCamp

COLLEGIATE AFFILIATIONS

- **Institute of Computer Engineers of the Philippines Student Edition (ICpEP.SE)** — Colegio de Muntinlupa Chapter (Active Member, 2022 – Present)

ONLINE PRESENCE & LINKS

- **Portfolio:** aera0908.github.io
- **GitHub:** github.com/Aera0908
- **LinkedIn:** linkedin.com/in/aira-josh-ynte
- **LeetCode:** leetcode.com/u/Aera0908
- **Fehuvia Project:** fehuvia.app
- **Safehouse Project:** safehouse-inky.vercel.app